12 Physics 2017
Programme

WEEK	CONTENT	TEXT	PROBLEM SOLVING TEXT	PROBLEM SOLVING MANUAL	PRACTICALS	ASSESSMENT
Spring Term 3-4	**Motion** Incline planes. Projectiles - vertical + horizontal components of velocity. Effect of air resistance. Assignment 1 - Projectile Motion.	2.1 2.2, 2.3	2.1 2.2, 2.3	Set 3		
5-6	**Circular Motion** Uniform horizontal motion - centripetal force + acceleration. Motion while banking and conical pendulums. Vertical motion - roller coasters.	2.4, 2.5 2.6	2.4, 2.5 2.6	Set 4	Expt 4.1 - Circular motion	Prac Collection 1 Test - Projectiles and Circular Motion
7-8	**Gravitational Fields** Newton's Law of Gravitation. Gravitational field strength. Work done by gravitational field. Kepler's laws of planetary motion. Satellite motion. Assignment 2 - Circular Motion and Gravitation.	1.1 1.2 1.3 2.6 2.7	1.1 1.2 1.3 2.6 2.7	Set 5		
9	**Moments (Torque)** Turning effect of a force. Centre of mass and stability.	3.1	3.1	Set 2	Expt 2.1 - Parallel forces	
Summer Term 1	Conditions for equilibrium. Forces in structures. Assignment 3 - Moments and Equilibrium.	3.2, 3.3	3.2, 3.3	Set 2	Expt 2.2 - Forces, torques in equilibrium Expt 2.3 - Non-parallel forces	Prac Collection 2 Test - Gravity and Torque

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2-5	**Magnetic fields** Fields around magnets, Earth, moving charges, current-carrying wires. Force on current-carrying wires. DC Motors - construction, torque.	5.1 5.3	5.1 5.3	Set 6	Investigation: Earth's magnetic field Investigation: Force on a wire	Prac Collection 3
6	Forces on isolated charges. Mass spectrometer, particle accelerators. Assignment 4 - Electromagnetism.	5.2	5.2	Set 10, 11		Test: Electromagnetism
7-10	**Electromagnetic Induction** Induced emf produced by the relative motion of a straight conductor in a magnetic field. Changing magnetic flux induces a potential difference - Faraday's Law. Lenz's Law of electromagnetic induction.	6.1 6.2 6.3	6.1 6.2 6.3	Set 7 Set 8	Expt 8.1 - Transformers Expt 8.4 - Back EMF	
Autumn Term 1-2	Transformers, AC and DC generators, induction hotplates, regenerative braking systems, AC power distribution. Assignment 5 - Induced EMF.	6.4	6.4	Set 8		Test - Electromagnetic Induction
3	**Revision**					
4-5	**Semester 1 Examinations**					
6-7	**Wave Particle Duality** Light exhibits wave properties - reflection, refraction, dispersion, diffraction, interference, polarisation. Electromagnetic waves are transverse waves made up of oscillating electric and magnetic fields. Evidence for the dual nature of light.	7.1, 7.2 7.3 7.4	7.1, 7.2 7.3 7.4		Investigation: Behaviour of light	

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8-9	**Quantum Theory** States of matter and energy are quantised into discrete values. Electromagnetic radiation is emitted or absorbed in discrete packets called photons. Black body radiation and the photoelectric effect are explained using the concept of light quanta. Absorption and emission spectra. Bohr model of the hydrogen atom. On the atomic level, energy and matter exhibit the characteristics of both waves and particles. Applications of quantum physics, including lasers, photovoltaic cells. Assignment 6 - Waves and Quantum Theory.	7.4 7.5 7.6	7.4 7.5 7.6	Set 12 Set 13	Expt 12.5 - Line spectra	Evaluation and analysis - Quantum theory and relativity Test - Wave Particle Duality and Quantum Theory
10	**Special Relativity** Observations of objects travelling at very high speeds cannot be explained by Newtonian physics. e.g. dilated half-life of high-speed muons, the momentum of high-speed particles in particle accelerators. Einstein's special theory of relativity - used for velocities approaching the speed of light.	8.1 8.2	8.1 8.2	Set 14		
Winter Term 1-3	Motion can only be measured relative to an observer - length and time are relative quantities that depend on the observer's frame of reference . Mass-energy equivalence.	8.3, 8.4 8.5	8.3, 8.4 8.5			Test - Special Relativity
4-5	**The Standard Model** Standard Model is based on the premise that all matter in the universe is made up from elementary matter particles called quarks and leptons. Standard Model explains three of the four fundamental forces (strong, weak and electromagnetic forces) in terms of gauge bosons. Conservation of lepton and baryon numbers in reactions.	9.1 9.2	9.1 9.2	Set 15		

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6	LIFE Week					
7-8	High-energy particle accelerators. Expansion of the universe - Hubble's law, red shift, Big Bang Theory. Standard Model is used to describe the evolution of forces and the creation of matter in the Big Bang Theory. Assignment 7 - Relativity and the Standard Model.	9.3 9.4	9.3 9.4			Test - Standard Model
9	**Revision**					
10	**Semester 2 Examination**					

References

Text: Physics Western Australia 12 (Pearson)
Manual: Exploring Physics – Year 12 (Science Teachers' Association of WA)
Practicals: Exploring Physics – Year 12 (Science Teachers' Association of WA)